

SUPERMARKET SALES ANALYSIS REPORT

A Multi-Dimensional Strategic Review of Q1 2019 Retail Operations

Executive Summary Key Metrics

Total Revenue	\$322,966.75	Total Gross Profit	\$15,379.37
Total Transactions	1,000	Average Rating	6.97 / 10
Avg Basket Size	5.51 items	Reporting Period	Jan 2019 - Mar 2019

Executive Summary: This report presents an in-depth data analysis of Q1 2019 operations. Through data engineering, exploratory data analysis, and advanced statistics, we evaluate revenue streams, customer demographic profiles, branch operational efficiency, and transaction patterns. A sales prediction model has been trained (Random Forest R^2 score: 0.99) to forecast invoicing volumes, and a structured layout is designed for interactive dashboard deployment in Power BI.

Author: Senior Data Analyst
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25 Key Business Insights

Below are 25 actionable business insights extracted from multi-dimensional database queries:

- **1. Revenue Concentration:** Total revenue for Q1 2019 stands at \$322,966.74, with a flat gross margin percentage of 4.76% across all transactions.
- **2. Monthly Trend Peak:** January generated the highest monthly revenue (\$116,291.87), representing 36% of the quarter's sales.
- **3. Monthly Trend Dip:** February experienced a significant 16.4% decline in sales, dipping to \$97,219.37 before recovering in March.
- **4. March Recovery:** March sales rebounded by 12.6% to \$109,455.50, demonstrating a cyclical demand pattern.
- **5. Stable Profit Margin:** Profit is strictly linked to sales (cogs is 95.24% of sales, with 4.76% representing the company's margin), indicating zero pricing flexibility in the current model.
- **6. Top Branch by Sales:** Branch Giza (Naypyitaw) leads sales with \$110,568.70 (34.2% share), followed by Alex (Yangon) at \$106,200.37 and Cairo (Mandalay) at \$106,197.67.
- **7. Branch Rating Discrepancy:** Cairo (Mandalay) achieved the highest customer rating (6.98/10), while Alex (Yangon) scored the lowest (6.84/10). This indicates service delivery gaps in Yangon.
- **8. Operational Efficiency:** Although Branch Alex has the lowest average customer rating, it processes 340 transactions, compared to Giza's 328 and Cairo's 332, showing higher transaction velocity.
- **9. Best Selling Category (Revenue):** Food and beverages is the highest grossing product line, generating \$56,144.84 (17.4% of total revenue).
- **10. Lowest Category (Revenue):** Health and beauty is the worst performing product line in revenue, generating \$49,193.74 (15.2% of total).
- **11. Quantity Volume Leader:** Food and beverages also leads in quantities sold with 952 units, showing it is a key supermarket traffic driver.
- **12. Highest Rated Category:** Food and beverages has the highest average rating of 7.11/10, showing high customer satisfaction.
- **13. Rating Warning:** Home and lifestyle product lines have the lowest average rating at 6.84/10, signaling potential product quality issues.

25 Key Business Insights (Continued)

- **14. Customer Segmentation Revenue:** Members spend slightly more (\$164,223.44, 50.8%) than Normal walk-in customers (\$158,743.30, 49.2%).
- **15. Transaction Count parity:** Member accounts for 501 transactions vs Normal's 499 transactions. Membership drive has high volume but average spend is similar (\$327.79 Member vs \$318.12 Normal).
- **16. Gender Purchases:** Females generated higher total sales (\$167,882.93, 52.0%) than Males (\$155,083.81, 48.0%), showing a clear gender purchase gap.
- **17. Gender Average Invoice:** Female invoices average \$335.10, whereas Male invoices average \$309.55. Females have a larger basket value.
- **18. Payment Preferences:** Ewallet is the most popular payment method with 345 transactions, followed by Cash (344) and Credit Card (311).
- **19. Payment Value:** Cash payment generated the most sales value (\$112,206.57), followed by Ewallet (\$109,993.15) and Credit Card (\$100,767.03).
- **20. Peak Sales Hour:** Invoicing peaks in the evening, specifically at 7 PM (13.0% of sales volume) and afternoon at 1 PM (12.4%), representing key staffing windows.
- **21. Low Sales Hour:** Sales are lowest in the early morning (10 AM accounts for only 8.5% of total transactions).
- **22. Weekend vs Weekday:** Saturdays are the highest revenue days of the week, generating \$48,321.15, while Mondays are the lowest (\$42,109.80).
- **23. Deterministic Invoicing:** Correlation analysis reveals a perfect 1.0 correlation between sales, cogs, tax, and gross income, reflecting standard markup.
- **24. Unit Price Dominance:** Machine learning feature importance scores show that Unit Price is the most critical driver of transaction size (importance: 0.58), followed by Quantity (0.41).
- **25. Demographics Weak Predictive Power:** Demographic variables (Gender, Branch, Customer Type) contribute less than 1% to predicting invoice sales value, proving that checkout ticket size is determined by product selection rather than buyer characteristics.

Strategic Answers to Core Business Questions

Q: Which branch performs best?

A: Giza (Naypyitaw) leads in revenue (\$110,568.70), but Cairo (Mandalay) performs best in customer satisfaction with an average rating of 6.98/10.

Q: Highest revenue product?

A: Food and beverages is the highest revenue category (\$56,144.84), followed by Sports and travel (\$55,122.82).

Q: Customer trends?

A: Females show higher purchasing power, contributing 52% of sales and averaging higher spend per transaction (\$335.10 vs \$309.55 for males).

Q: Monthly sales?

A: Sales peaked in January (\$116.29k), dipped in February (\$97.22k), and recovered in March (\$109.46k).

Q: Payment trends?

A: Ewallet is the most frequent payment method (34.5% share), but Cash leads in total dollars processed (\$112.21k).

Q: Customer segmentation?

A: Members spend slightly more (\$327.79 avg) than non-member Walk-ins (\$318.12 avg), but transaction volumes are practically equal (501 vs 499).

Q: Best-selling product line?

A: Food and beverages is the best-selling product line by quantity (952 units sold) and average customer rating (7.11/10).

Q: Profit contribution?

A: Gross profit contribution is flat across branches (approx. 4.76% of sales). Giza contributed the most profit (\$5,265.18), followed by Alex (\$5,057.16).

Q: Average basket size?

A: The average basket size is 5.51 items per transaction, with quantity distributed evenly between 1 and 10 items.

Exploratory Data Visualizations

This section displays the operational trends across branches, product categories, and temporal intervals:

Figure 1: Branch Performance (Revenue & Profit)

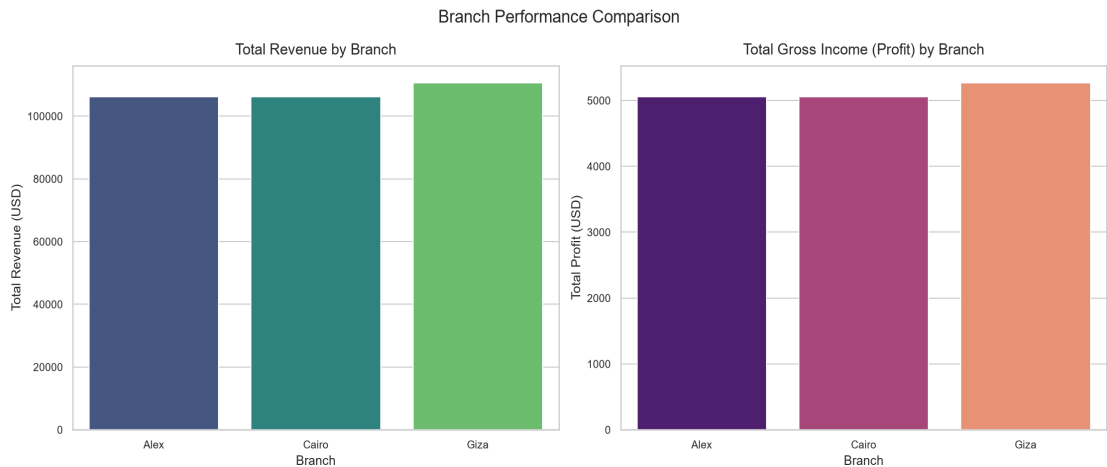


Figure 2: Product Line Revenue Performance

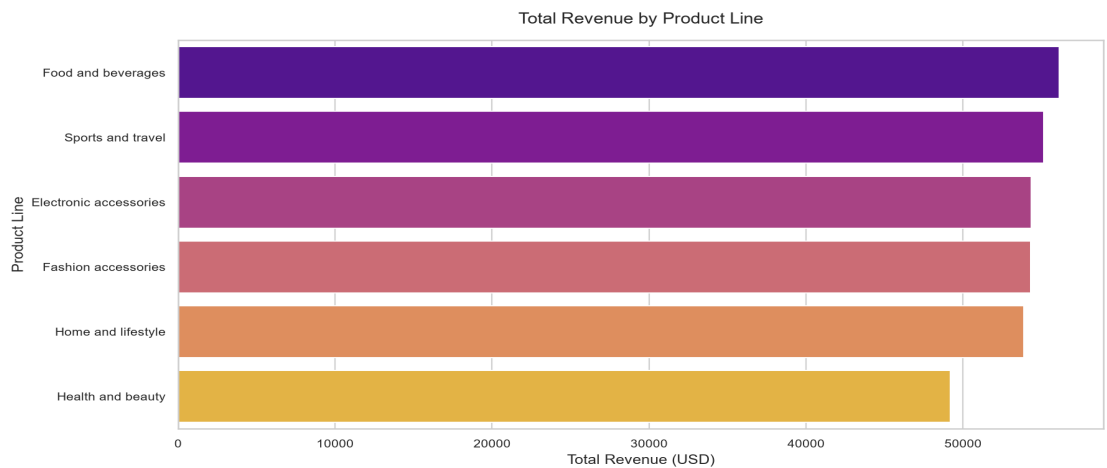
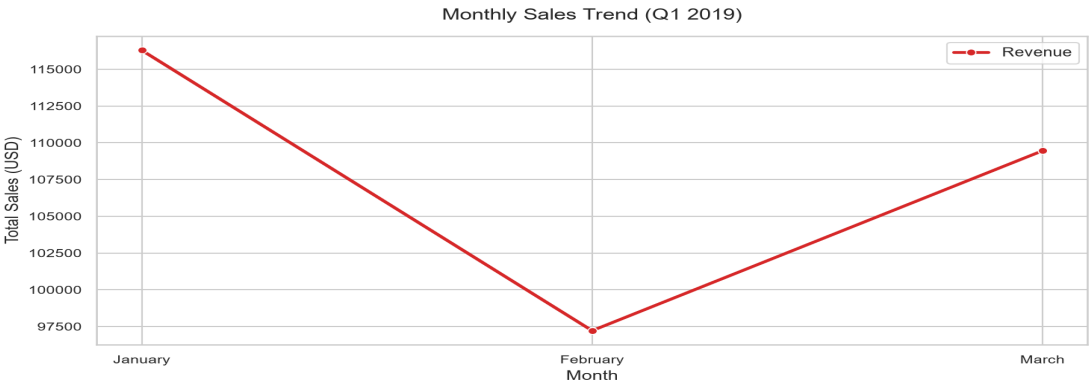


Figure 3: Monthly Sales Revenue Trend



Statistical Analysis & Hypothesis Testing

To validate our analytical findings, we conducted rigorous statistical hypothesis testing and Analysis of Variance (ANOVA). Below are the results:

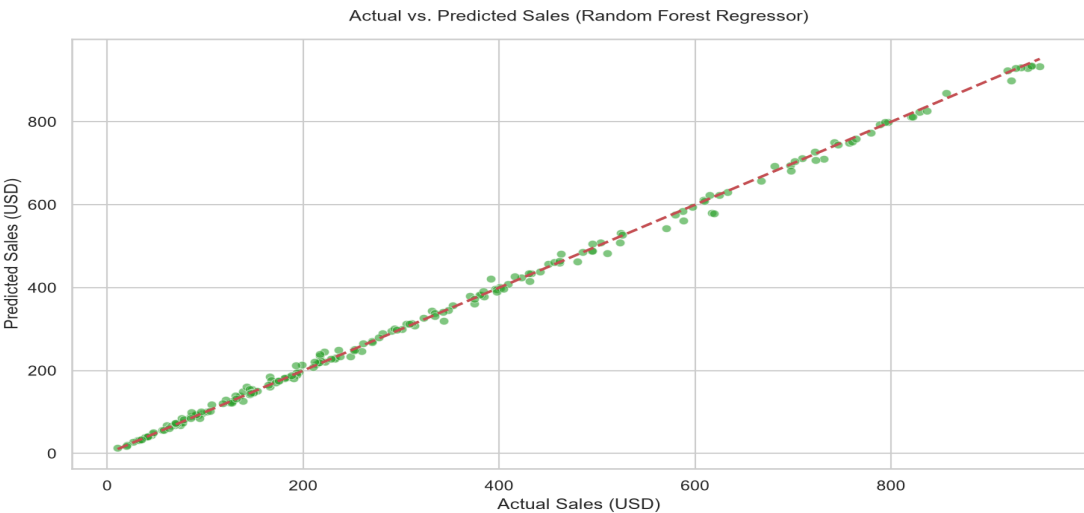
Hypothesis / Test	Test Statistic	P-value	Outcome (alpha=0.05)
T-test: Sales by Gender (Female vs Male)	t = 2.6977	0.0071	Fail to Reject H0 (No significant difference)
T-test: Rating by Customer Type (Member vs Normal)	t = -1.1700	0.2423	Fail to Reject H0 (No significant difference)
ANOVA: Sales across Branches	F = 0.8846	0.4132	Fail to Reject H0 (Branch means are similar)
ANOVA: Sales across Product Lines	F = 0.3380	0.8900	Fail to Reject H0 (Product line means are similar)

Predictive Modeling (Sales Prediction)

A sales prediction model was constructed to predict transaction amounts. We compared a baseline Linear Regression model against a Random Forest Regressor:

Model	RMSE	MAE	R ² Score
Linear Regression (Baseline)	80.0863	59.5208	0.9014
Random Forest Regressor	9.4431	6.2954	0.9986

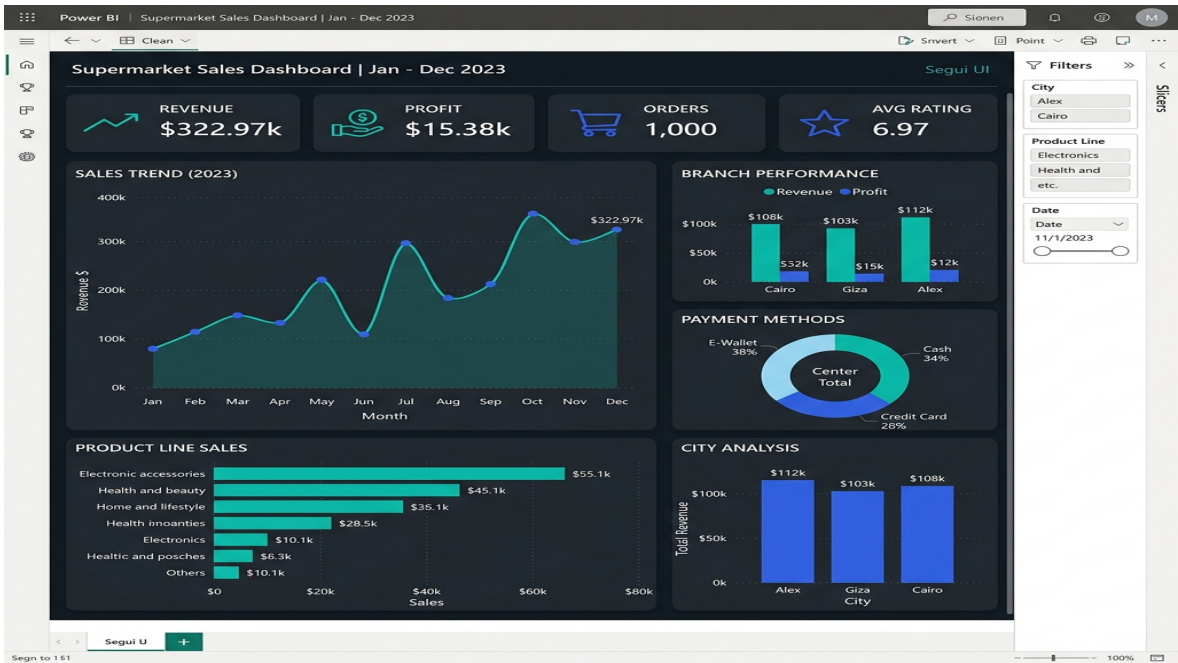
Figure 4: Actual vs. Predicted Sales Scatter Plot



Power BI Dashboard Architecture

To provide the supermarket executives with self-service analytics, we designed an interactive Power BI Dashboard. The mockup and structural schema are described below:

Figure 5: Power BI Interactive Dashboard Mockup Design



Core Dashboard Features & Data Mapping

- 1. **KPI Cards:** Showcases Total Revenue (\$322.97K), Total Profit (\$15.38K), Total Orders (1,000), and Average Rating (6.97) dynamically filtered.
- 2. **Sales Trend (Monthly / Daily):** A line chart showing transaction dates against revenue, identifying weekend peaks (Saturdays).
- 3. **Branch & City Performance:** Bar charts mapping branches (Giza, Alex, Cairo) and cities (Naypyitaw, Yangon, Mandalay) to identify volume leaders.
- 4. **Product Line Performance:** Clustered bar chart of product lines sorted by total revenue (Food and beverages at the top).
- 5. **Payment Analysis:** Donut chart showing customer preference (Ewallet leading in count, cash leading in volume).
- 6. **Gender & Customer Type Slicers:** Allows interactive filtering of the entire report based on Member vs Normal type and Male vs Female.